Zero-Day Detection under HAI Plant-Version Shift

Source-only PCA, DeepSVDD and VAE with frozen SPOT or adaptive DSPOT | Zhang and Wang, Group 88, SSY340

EXPERIMENTAL SETUP

Data: 1,608,008 real 1-Hz rows; 59 shared tags filtered to 41 source-nonconstant tags; 102 documented attacks (52 in 23.05).

Protocol: three seeds (11, 23, 37); normal 21.03-only training and calibration; chronological 23.05 replay.

Models: shared bias-free CNN trunk for DeepSVDD/VAE, standard VAE objective, and PCA reconstruction. Causal 60 s windows, 10 s stride, max sensor fusion.

Metrics: raw point metrics without point adjustment; event detection; eTaPR (theta_p=0.5, theta_r=0.1; SAC 2022, not CIKM 2019); PR-AUC; delay; attribution hit@3.

KEY FINDINGS

- VAE achieved the highest mean 23.05 PR-AUC (0.232 ± 0.002). Frozen and adaptive PR-AUC are identical because both arms threshold the same continuous scores.

- For VAE, frozen event recall moved from 0.560 on 21.03 to 0.154 on 23.05; DSPOT changed cross-domain recall to 0.115 (gain -0.038; recovery ROM -0.095) and false alarms from 16.41 to 5.47 per day.

- The highest raw frozen 23.05 event recall belonged to PCA and DeepSVDD (1.000 ± 0.000); the highest raw adaptive recall belonged to PCA and DeepSVDD (1.000 ± 0.000). Raw recall is not a stand-alone measure of useful detection.

INTERPRETATION AND LIMITS

DSPOT adapts the decision layer, not the representation. Equal PR-AUC across arms is therefore expected: both use identical continuous scores. A thresholded-metric gain is not evidence of representation improvement. The source alarm target is not a shifted-domain guarantee. Attacks lacking a retained documented public target are excluded from attribution-hit denominators, never treated as attribution failures. Three-seed SDs are descriptive; delays condition on detected attacks.

23.05 THREE-SEED MEANS

Model	Event F	Event A	Gain	PR-AUC
PCA	1.000	1.000	+0.000	0.110
DeepSVDD	1.000	1.000	+0.000	0.091
VAE	0.154	0.115	-0.038	0.232

F = frozen SPOT; A = adaptive DSPOT; Event = documented-attack detection rate. Gain = adaptive minus frozen. PR-AUC is arm-invariant by construction.

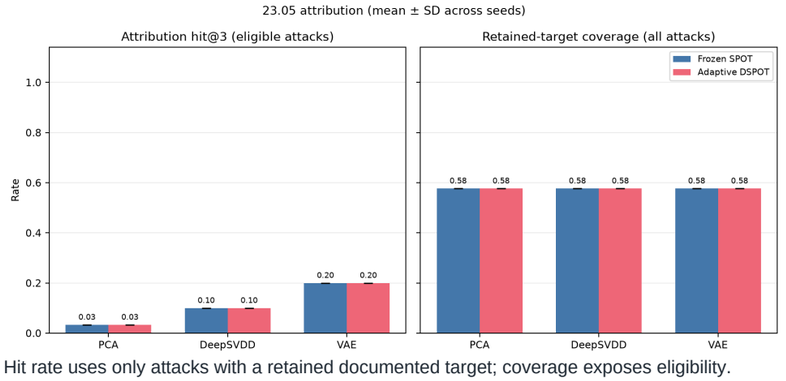
RECOVERY AND OPERATING COST

PCA and DeepSVDD had the largest signed mean DSPOT event-detection gain (+0.000; ratio-of-means recovery undefined). A positive gain is recovery, whereas a negative gain is an additional loss.

Observed 23.05 false alarms spanned 5.470–8296.588 per day across runs; the source-normal calibration target was 10 per day. Near-continuous-alarm failure screen (at least 99.9% event recall and at least 90% normal-window false alarms): PCA / frozen (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%); PCA / adaptive (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%); DeepSVDD / frozen (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%); DeepSVDD / adaptive (mean recall 1.000, mean FA/day 8296.59, normal-window alarm fraction 96.0%). These high-recall conditions are alarm-collapse failures, not evidence of useful detection or DSPOT recovery.

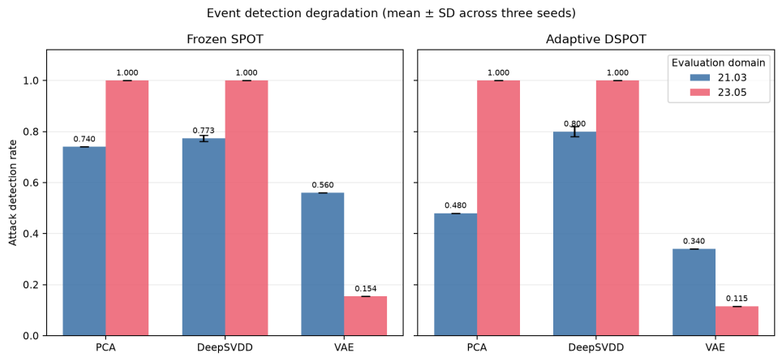
Of the 52 unique 23.05 attacks, 30 retained a documented target in the 41-tag source-nonconstant modeling schema and 22 did not. The latter are excluded from hit-rate denominators, not counted as attribution failures.

ATTRIBUTION

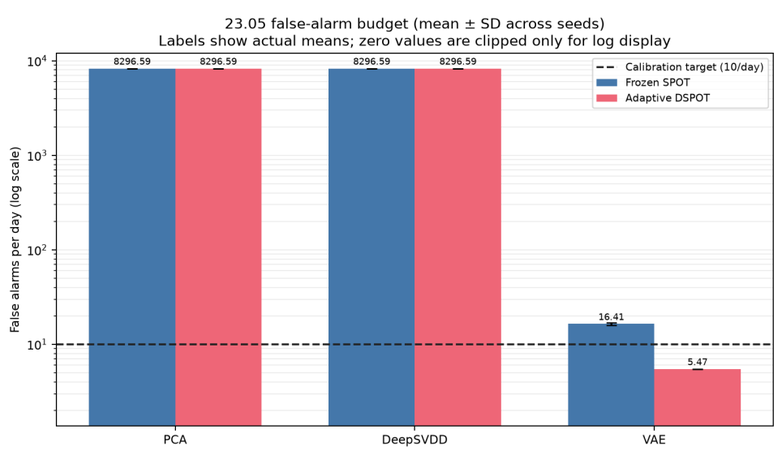


Hit rate uses only attacks with a retained documented target; coverage exposes eligibility.

EVENT DEGRADATION AND ADAPTATION



FALSE-ALARM BUDGET



Bars show three-seed means with SD; the dashed line is the normal-source calibration objective.

References: HAI/CSET 2020, 2021; Deep SVDD/ICML 2018; Rethinking/2020; SPOT/KDD 2017; TaPR/CIKM 2019; eTaPR/SAC 2022; Kim et al./AAAI 2022.

Full numerical results and per-attack failures: final_report.pdf